

Agenda

1.) Announcements

- No discussion quiz this week
- We'll have your exams graded next Wednesday
- I will not have office hours today

2.) Section 11.10

Very Quick Review of Power Series (11.8, 11.9, 11.10, 11.11)

done

↑
we are
here

next week

- 1.) Power series are what we get when we add variables to infinite series:

$$\sum_{n=0}^{\infty} a_n (x-c)^n \text{ is a power series centered at } x=c.$$

- 2.) Instead of asking "does it converge/diverge", we ask: "For what values of x does the power series converge?"

- 3.) We use the radius of convergence and interval of convergence to answer this question.

Possible ROC: 0 ∞ $0 < R < \infty$

Possible IOC: $\{c\}$ $(-\infty, \infty)$ $(c-r, c+r)$
 $[c-r, c+r)$
 $(c-r, c+r]$
 $[c-r, c+r]$ } test endpoints

- 4.) We can use the fact $\sum_{n=0}^{\infty} ar^n$ converges to $\frac{a}{1-r}$ when $|r| < 1$ to find power series representations of functions that are similar to $\frac{1}{1-x}$ (or its derivative, or its integral)

- 5.) There's a formula of sorts for finding power series:

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!} (x-c)^n$$

These are called Maclaurin series when $c=0$.

Otherwise, they are called Taylor series.

Maclaurin Series and their Radius of Convergence

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, \text{ with a radius of convergence } R = 1.$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \text{ with a radius of convergence } R = \infty.$$

$$\sin(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}, \text{ with a radius of convergence } R = \infty.$$

$$\cos(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}, \text{ with a radius of convergence } R = \infty.$$

$$\tan^{-1}(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}, \text{ with a radius of convergence } R = 1.$$

$$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}, \text{ with a radius of convergence } R = 1.$$

$$(1+x)^k = \sum_{n=0}^{\infty} \binom{k}{n} x^n, \text{ with a radius of convergence } R = 1. \quad \text{Ignore this for now.}$$

Activity: Evaluate each of the following power series:

$$1.) \sum_{n=0}^{\infty} \frac{2^{2n+1}}{5^n} = \sum_{n=0}^{\infty} 2 \left(\frac{4}{5}\right)^n = \frac{2}{1-\frac{4}{5}} = 10$$

$$2.) \sum_{n=0}^{\infty} \frac{(\ln 2)^n}{n!} = e^{\ln 2} = 2$$

Allows us to
find exact values
of particular infinite series

$$3.) \sum_{n=0}^{\infty} (-1)^n \frac{\pi^{2n+1}}{2^{2n+1} (2n+1)!} = \sum_{n=0}^{\infty} (-1)^n \frac{\left(\frac{\pi}{2}\right)^{2n+1}}{(2n+1)!} = \sin\left(\frac{\pi}{2}\right) = 1$$

$$4.) \sum_{n=0}^{\infty} (-1)^n \frac{1}{2^{2n+1} (2n+1)} = \sum_{n=0}^{\infty} (-1)^n \frac{\left(\frac{1}{2}\right)^{2n+1}}{2n+1} = \arctan\left(\frac{1}{2}\right)$$

Calculus with Taylor Series

Example: Use the Maclaurin Series for $f(x) = e^x$ to show
that $f'(x) = e^x$

$$f(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots$$

$$f'(x) = \sum_{n=0}^{\infty} \frac{n x^{n-1}}{n!} = 0 + 1 + \frac{2x}{2} + \frac{3x^2}{6} + \dots = 1 + x + \frac{x^2}{2} + \dots$$

$$= \sum_{n=0}^{\infty} \frac{\cancel{n} x^{n-1}}{\cancel{n} (n-1)!} = \sum_{n=0}^{\infty} \frac{x^{n-1}}{(n-1)!} = \sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x$$

Example: Find a power series representation of $g(x) = xe^{x^2}$

$$f(x) = e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$g(x) = xe^{x^2} = x \sum_{n=0}^{\infty} \frac{(x^2)^n}{n!} = x \sum_{n=0}^{\infty} \frac{x^{2n}}{n!} = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{n!}$$

Activity: Use power series to integrate: $\int \sin(x^2) dx$

$$f(x) = \sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

$$g(x) = \sum_{n=0}^{\infty} (-1)^n \frac{(x^2)^{2n+1}}{(2n+1)!} = \sum_{n=0}^{\infty} (-1)^n \frac{x^{4n+2}}{(2n+1)!}$$

$$\int \sin(x^2) dx = \int \left[\sum_{n=0}^{\infty} (-1)^n \frac{x^{4n+2}}{(2n+1)!} \right] dx = C + \sum_{n=0}^{\infty} (-1)^n \frac{x^{4n+3}}{(4n+3)(2n+1)!}$$

Example: Use power series to evaluate $\lim_{x \rightarrow 0} \frac{\sin x}{x}$

$$\lim_{x \rightarrow 0} \frac{\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}}{x} = \lim_{x \rightarrow 0} \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n+1)!} = \lim_{x \rightarrow 0} \left(1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \dots \right) = 1$$

Activity: Use power series to evaluate $\lim_{x \rightarrow 0} \frac{\cos x - 1}{x^2}$

$$\lim_{x \rightarrow 0} \frac{\left(\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} \right) - 1}{x^2} = \lim_{x \rightarrow 0} \frac{\sum_{n=1}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}}{x^2} = \lim_{x \rightarrow 0} \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n-2}}{(2n)!}$$

$$= \lim_{x \rightarrow 0} \left(-\frac{1}{2!} + \frac{x^2}{4!} - \frac{x^4}{6!} + \dots \right)$$

$$= -1/2$$

